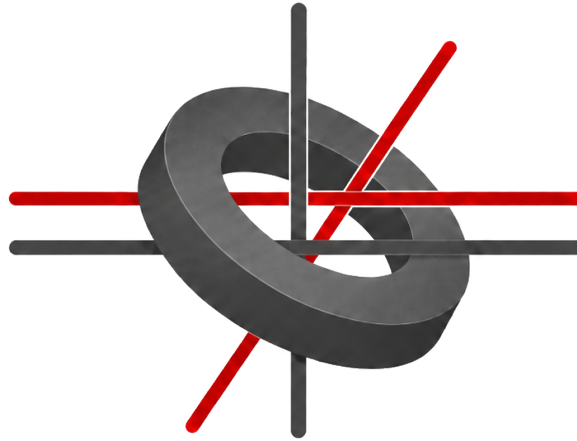




Final Degree Project Documentation in Computer Engineering in the AI Era

An unofficial, academic and practical guide for TFG documentation at USAL

Alvaro Lozano Murciego

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This English version is a structural placeholder. The first complete version of the book is maintained in Spanish.

Note

Translation pending. Always use the Spanish version as the source of truth for now.

The book is an unofficial, opinionated and practical guide for documenting a Final Degree Project in Computer Engineering at the University of Salamanca.

Part I

TFG Regulations

TFG REGULATIONS

Translation pending. Use the Spanish version as the source of truth.

1.1 USAL Regulations

Translation pending. Use the Spanish version as the source of truth.

1.2 Faculty of Sciences Regulations

Translation pending. Use the Spanish version as the source of truth.

1.3 Department Regulations

Translation pending. Use the Spanish version as the source of truth.

1.4 Professional Standards

Translation pending. Use the Spanish version as the source of truth.

Part II

TFG Documentation

TFG DOCUMENTATION

Translation pending. Use the Spanish version as the source of truth.

2.1 Style Rules

Translation pending. Use the Spanish version as the source of truth.

2.2 Citations and Bibliography

Translation pending. Use the Spanish version as the source of truth.

This book uses an APA-like author-year citation format because it fits an academic guide that combines regulations, web documentation, manuals and teaching references. If a supervisor, committee or official template requires IEEE or another style, that requirement takes precedence.

2.3 Current Structure

Translation pending. Use the Spanish version as the source of truth.

2.4 Old ITIS/PFC Structure

Translation pending. Use the Spanish version as the source of truth.

2.5 Old vs Current Mapping

Translation pending. Use the Spanish version as the source of truth.

THE REPORT

The report is the document that the assessment committee should be able to read from start to finish in order to understand the TFG. It should not contain every technical detail, but it must contain the important decisions: what problem is addressed, why it matters, what has been built or studied, how it has been validated and what can be concluded.

The DIA v2 guide presents the report as the document that connects the rest of the TFG documentation and justifies the adopted solutions (Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024). That is the central idea: the report does not replace the annexes; it guides them. The reader should be able to follow the main reasoning without reading every detailed requirement, diagram or test log.

3.1 What should accompany the report

Before the technical chapters, the documentation should include several formal elements:

- **Cover page:** clear title, author, defense date and supervisor or supervisors. The title should be meaningful and avoid unexplained acronyms.
- **Supervisor consent form:** include the current official model when the call or Department requires it.
- **Resumen and abstract:** self-contained synthesis in Spanish and English, usually between 200 and 500 words. It should include the problem, methodology, tools, results and conclusions.
- **Glossary:** useful when the report uses domain terms, non-standard abbreviations or acronyms.
- **Indexes:** report table of contents, annex index and tables/figures indexes when used.

Figures and tables need captions and textual references. External resources must be cited. Mandatory requirements should use clear obligation wording: “must”, “shall” or equivalent. Optional improvements should be marked as desirable rather than compulsory.

3.2 How to use the sections

The following sections should remain connected:

- Objectives explain what the TFG tries to achieve.
- Background justifies why the problem deserves attention.
- Methods, tools and models explain the approach.
- Requirements, constraints and scope define the limits.
- The proposed solution shows what has been produced.
- Management, testing and risks show control of the process.

- Conclusions return to the objectives and assess the result.

When an explanation becomes too detailed for the report, move it to an annex and leave a summary and cross-reference here.

3.3 Introduction

The introduction opens the report and should orient the committee before entering technical detail. According to the DIA v2 guide, it should present the problem domain, its context, possible solutions, the motivation for the work and the overall structure of the documentation ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

A useful introduction answers these questions early:

- What domain does the TFG address?
- What problem, need or opportunity has been identified?
- Why does it make sense as a Computer Engineering TFG?
- What general approach has been followed?
- How is the report organized?

Do not duplicate the abstract. The abstract condenses the whole work; the introduction prepares the reader. It may mention general alternatives, but it should not reveal the whole design or become a tutorial on the technologies used.

3.3.1 Recommended structure

A solid structure is:

1. Present the context and application area.
2. Explain the problem with concrete examples.
3. Justify the academic, technical or social motivation.
4. Anticipate the type of solution or study performed.
5. Close with a short map of the document.

Keep the wording precise. Avoid generic statements if they do not support the project. It is better to explain what real limitation exists, who it affects and why your work provides an assessable response.

3.4 Purpose and objectives

This section defines what the TFG aims to achieve. The DIA v2 guide asks for functional and personal objectives, with one main objective and several achievable, reasonable and measurable subobjectives ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

The **main objective** should express the central contribution: building a system, designing a method, evaluating a technique, developing a prototype, automating a process, comparing alternatives or generating applicable knowledge.

The **subobjectives** split that promise into verifiable results. They are often clearer when written with action verbs:

- Analyze the problem and its constraints.
- Design an architecture or solution model.
- Implement a working prototype.

- Evaluate the solution through metrics, tests or user validation.
- Document the process so that it is reproducible.

3.4.1 Functional and personal objectives

Functional objectives describe what the work should produce. Personal objectives describe relevant learning by the author, but only when connected to the TFG. “Learning a technology” is not enough; it is better to state that a specific technology will be applied and evaluated to solve a defined need.

3.4.2 Quality criteria

An objective is well written if the final report can answer:

- Has it been achieved?
- With what evidence?
- What limitation remains?
- Which technical decision helped achieve it?

Do not turn this section into a wish list. If a subobjective cannot be verified in the report, annexes, repository or tests, it probably needs rewriting.

3.5 Background and current situation

This block has two functions: explaining the background of the problem and describing the current situation. The DIA v2 guide separates both points: background gives the theoretical or empirical basis of the domain, while the current situation presents a state of the art, market study or existing solutions depending on the TFG ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

3.5.1 Background

The background should explain the non-computing domain when needed. If the TFG deals with education, health, business management, heritage, sports or scientific research, this section should give the committee the minimum basis needed to understand the problem.

Include:

- essential domain concepts;
- relevant previous studies or works;
- limitations in previous solutions;
- needs of users, organizations or processes;
- company context, only when applicable and allowed.

Theory should serve the TFG. There is no need to write a manual for every technology used; explain the prior knowledge that the committee needs in order to understand your decisions.

3.5.2 Current situation

Compare existing alternatives: similar applications, reference tools, commercial products, repositories, papers, regulations or common practices. If technological evolution matters, describe it briefly, but avoid a long history with no connection to your solution.

This section should rely on references. Do not simply claim that no adequate solution exists. Show what has been reviewed, what each alternative offers and why your chosen path makes sense.

3.6 Methods, tools, and models

The DIA v2 guide groups here standards, references, methods, tools, models, prototypes and metrics ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). The goal is not to list everything installed, but to explain the technical approach and justify decisions that affect the result.

3.6.1 Methods

Describe the technical methodology: unified process, agile development, iterative prototyping, experimental methodology, systematic review, user-centered design or another approach. If you rely on known concepts, cite them and summarize only what is needed.

The method should explain how the work moved from problem to solution:

- project phases;
- decision criteria;
- validation strategy;
- relation between methodology and planning.

3.6.2 Tools

Group tools by function:

- **Implementation:** languages, frameworks, libraries, databases, APIs, cloud services, devices or models.
- **Methodology:** task management, version control, continuous integration, testing, design, prototyping or technical documentation.
- **TFG writing:** LaTeX, Markdown, Jupyter Book, bibliography managers, diagram tools or AI assistants when they materially influenced the work.

Do not explain standard tools if they add no value. Saying that a text editor was used is rarely relevant; explaining why a specific AI library, architecture or database was chosen is relevant.

3.6.3 Models, prototypes and metrics

Include models when they are part of the work: AI, mathematical, industrial, electronic, simulation, data or decision models. If there are wireframes, mock-ups or navigation diagrams, they should justify design and user experience.

Metrics are useful when they help assess the result: performance, accuracy, test coverage, user satisfaction, execution time, resource use, accessibility or documentation quality. A metric without interpretation proves little; always pair it with a criterion.

3.7 Requirements, scope, and constraints

The reader should understand what is inside and outside the TFG. The DIA v2 guide recommends summarizing initial requirements, hypotheses, constraints, scope, expected impact and the alternatives and feasibility study in the report (Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024).

3.7.1 Initial requirements

The report does not need to repeat the whole A1 annex. It should present a summary that makes the commitment clear:

- main functional requirements;
- relevant non-functional requirements;
- use cases, user stories or equivalent format;
- acceptance criteria;
- data, security, performance, accessibility, deployment or maintenance constraints.

Mandatory requirements should be written as commitments: “the system must...”, “the application shall...”. Use “should” for desirable improvements outside the minimum scope.

3.7.2 Hypotheses, constraints and scope

Constraints are not excuses; they are design conditions. They may come from available time, data, regulations, privacy, hardware, budget, APIs, prior knowledge, integration needs or limited access to users.

The scope should make clear:

- what the TFG includes;
- what is outside the work;
- what is left as future work;
- what level of completion is expected: prototype, usable product, research work, library, demonstrator or technical documentation.

It is also useful to explain the expected impact: which user, organization, process or sector benefits from the solution and how.

3.7.3 Alternatives and feasibility

The report should show that technologies were not chosen randomly. Compare alternatives for languages, frameworks, APIs, databases, models or services. The comparison can be brief, but it should justify the choice using criteria such as cost, maturity, documentation, performance, privacy, deployment, community, license or compatibility with requirements.

When applicable, add a qualitative assessment of costs and benefits. In a TFG, monetization is not always relevant, but effort, maintenance, reuse and real feasibility usually are.

3.8 Proposed solution

The proposed solution should technically explain the final product, method or result. The DIA v2 guide asks for functionalities, screenshots, diagrams and a way to test the development: deployment, executable, installer or demonstration video when applicable ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

This section must connect requirements and outcome. It is not enough to show screens; explain the relevant technical decisions, main components and how the solution responds to the objectives.

3.8.1 What should be clear

Include, depending on the TFG:

- overview of the solution;
- system architecture;
- main components and responsibilities;
- data or usage flow;
- implemented functionalities;
- screenshots or diagrams with explanation;
- technical decisions and discarded alternatives;
- repository, demo, deployment, installer or video link;
- tests or evidence of operation.

Diagrams should be referenced in the text and have captions. A screenshot without explanation has little value; a screenshot showing how a requirement is solved does.

3.8.2 From proposal to result

Explain how the solution evolved: problems found, decisions made, scope changes, limitations and particularities of the final version. This often separates a mature report from a superficial description.

If the solution includes AI, external models or APIs, document what the author controls and what depends on third-party services. If data is used, explain its origin, treatment and limitations. If there is deployment, state how it can be reproduced or tested.

3.8.3 Validation evidence

Close the solution with evidence: tests, metrics, user validation, comparison with requirements or result analysis. If the detail is large, keep the synthesis here and move full tables, cases or logs to the relevant annex.

3.9 Risks and management

This block should show that the project was managed with judgment. The DIA v2 guide distinguishes risk analysis, organization and project management ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). The report should not include a decorative Gantt chart; it should explain how work was organized and how problems were handled.

3.9.1 Risk analysis

Select risks that truly affect the TFG:

- **Acceptance:** users do not understand, need or adopt the solution.
- **Technology:** a library, API, model, device or architecture does not behave as expected.
- **Data:** lack of data, bias, poor quality, privacy or permissions.
- **Integration:** problems connecting the solution to a larger system.
- **Planning:** underestimated tasks, external dependencies or scope changes.

You may use a risk matrix, SWOT analysis or risk register. A minimal register can include number, description, priority, complexity and corrective measures. What matters is that each risk has a response: accept, reduce, avoid, transfer or monitor.

3.9.2 Work organization

Explain the real phases of the project and how the chosen methodology was applied. The path from proposal to final solution should be visible:

- initial analysis;
- design;
- implementation;
- testing;
- documentation review;
- defense preparation.

When architecture, components, protocols, entities, databases or algorithms are important, summarize the organization here and refer to the analysis and design annex for detail.

3.9.3 Project management

Include resources, timeline, milestones and deviations. If changes occurred, explain them naturally: a serious TFG is not one without problems, but one that identifies, manages and justifies adjustments.

The Gantt chart, if included, should tell the real story of the work. Connect planning, risks, decisions and final scope.

3.10 Conclusions

The conclusions close the report by returning to the objectives. The DIA v2 guide asks the author to make clear whether the objectives were achieved, what problems appeared, how they were solved, what technical and personal learning took place, what limitations remain and what future lines of work are possible ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

A serious conclusion does not merely state that the objectives were met. It should show it:

- objective;
- result obtained;
- evidence used to verify it;
- limitation found;
- possible improvement.

3.10.1 Results and learning

Summarize the main results without repeating the whole solution. Explain what worked, what did not go as far as expected and which decisions mattered most.

It is also reasonable to include professional and personal learning, as long as it is connected to the TFG: technologies learned, design judgment, time management, communication with users, data handling, documentation, testing or deployment.

3.10.2 Limitations and future work

Limitations are not failures; they are the honest boundary of the work. They may relate to time, data, performance, scale, validation, user access, privacy, generalization or maintenance.

Future work should be realistic and connected to what was actually done:

- extending a delimited functionality;
- improving performance or accessibility;
- validating with more users or data;
- integrating with another system;
- strengthening security or deployment;
- turning a prototype into a maintainable product.

3.10.3 Bibliography before submission

The DIA v2 guide recommends a standard citation format and mentions APA as a practical option ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). This book uses an APA-like author-year style. Before submission, check three things: every in-text citation appears in the bibliography, every bibliography entry has been cited at least once, and the sources are relevant to the TFG topic.

Translation pending. Use the Spanish version as the source of truth.

4.1 A1. System specifications

Translation pending. Use the Spanish version as the source of truth.

4.2 A2. System analysis and design

Translation pending. Use the Spanish version as the source of truth.

4.3 A3. Size and effort estimation

Translation pending. Use the Spanish version as the source of truth.

4.4 A4. Security plan

Translation pending. Use the Spanish version as the source of truth.

4.5 A5. Other annexes

Translation pending. Use the Spanish version as the source of truth.

Part III

Software Engineering

SOFTWARE ENGINEERING APPLIED TO THE TFG

Translation pending. Use the Spanish version as the source of truth.

5.1 RUP vs Agile

Translation pending. Use the Spanish version as the source of truth.

5.2 UML and Documentation as Code

Translation pending. Use the Spanish version as the source of truth.

5.3 Planning, Risks and Tests

Translation pending. Use the Spanish version as the source of truth.

Part IV

Tools

DOCUMENTATION TOOLS

Translation pending. Use the Spanish version as the source of truth.

6.1 Team Word

Translation pending. Use the Spanish version as the source of truth.

6.2 Team LaTeX

Translation pending. Use the Spanish version as the source of truth.

6.2.1 Starting templates in Overleaf

Public Overleaf templates can be useful as technical starting points for folder structure, LaTeX configuration, bibliography handling, and PDF layout ([Overleaf, 2026](#)). They are not official USAL or DIA templates. When using an external template, adapt it to the current TFG rules, review its license, remove institutional text that does not apply, and cite the original template.

Examples worth reviewing include the University of Seville TFG template ([Trinidad, 2026](#)), the ULL-ESIT-GII TFG report template ([Torres, 2026](#)), the University of Málaga TFG report template ([Hidalgo, 2026](#)), the University of Cádiz software-development TFG template ([Profesorado del Departamento de Ingeniería Informática de la Universidad de Cádiz, 2026](#)), and the UDIMA Computer Engineering TFG template ([Barrio, 2026](#)).

6.2.2 Overleaf from VS Code

If you want to write in VS Code while keeping the project in Overleaf, the cleanest route is usually Overleaf's Git integration, which lets you clone the project, work locally, and synchronize changes with `pull` and `push` ([Overleaf, 2026](#)). Another option is the community extension [Overleaf Workshop](#), which opens Overleaf/ShareLaTeX projects in VS Code and supports editing, compiling, previewing, and collaboration features ([iamhyc, 2026](#)).

Treat any extension-based workflow as an external tool: check authentication, permissions, project state, and conflicts before relying on it for final TFG work.

6.3 VS Code, AI Agents and Skills

Translation pending. Use the Spanish version as the source of truth.

Part V

Mistakes and FAQ

COMMON MISTAKES AND FAQ

Translation pending. Use the Spanish version as the source of truth.

7.1 Common Mistakes

Translation pending. Use the Spanish version as the source of truth.

7.2 FAQ

Translation pending. Use the Spanish version as the source of truth.

Part VI

Defense

TFG PRESENTATION AND DEFENSE

Translation pending. Use the Spanish version as the source of truth.

Part VII

Final Information

CREDITS

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LICENSES

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  year = {2026},  
  publisher = {Universidad de Salamanca},  
  note = {Unofficial guide, CC BY 4.0}  
}
```


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